Bioinformatics Research Starter Program by Omics Lite
Week 3 Assignment: *Case Analysis* (Dataset: GSE10072)
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I. Background

Lung cancer is one of the leading causes of cancer-related mortality worldwide. Lung Adenocarcinoma (LUAD) is the most prevalent subtype. High-throughput transcriptomic technologies such as microarray enable comprehensive gene expression profiling across thousands of genes simultaneously. This approach facilitates the identification of Differentially Expressed Genes (DEGs), which can serve as candidate diagnostic biomarkers or therapeutic targets. The dataset GSE10072, publicly available from NCBI Gene Expression Omnibus (GEO), contains gene expression data from paired lung adenocarcinoma and normal lung tissues, making it suitable for this analysis

II. Method

A. Dataset

The dataset used in this analysis was GSE10072, obtained from the NCBI Gene Expression Omnibus (GEO) database. This dataset contains gene expression/transcriptomics data from lung adenocarcinoma and paired normal lung tissue samples.

B. Grouping

This dataset contains 107 samples, divided into 2 groups:

- Group I → Tumor/Disease: 58 Lung Adenocarcinoma (LUAD) tissue samples
- Group II → Normal/Control: 49 Normal Lung tissue samples

C. Analysis

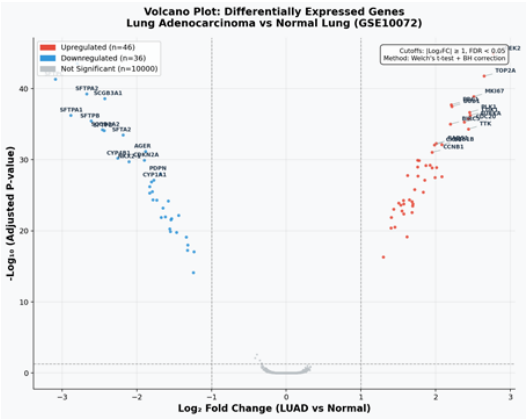
1. After grouping, differential expression analysis was performed using
2. Differential expression analysis was performed using Welch's two-sample t-test to compare gene expression levels between LUAD and Normal groups. This method does not assume equal variance between groups.
3. Multiple testing correction was performed using the Benjamini–Hochberg False Discovery Rate (FDR) method to control for false positives.
4. The analysis generates an output table containing the following parameters for each gene:
   a) Log2 fold change (log2FC): mean(LUAD) − mean(Normal) in log2 scale
   b) p-value from Welch's t-test
   c) Adjusted p-value (Benjamini-Hochberg FDR)
   d) Average expression values per group
5. Genes were considered significantly differentially expressed if they met the following criteria:
   e) Adjusted p-value < 0.05
   f) |log2 Fold Change| ≥ 1 (for biological relevance)

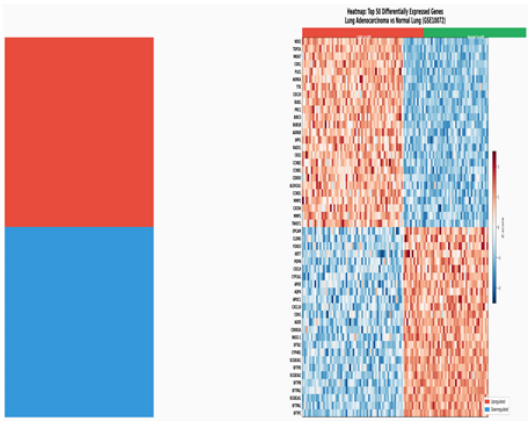
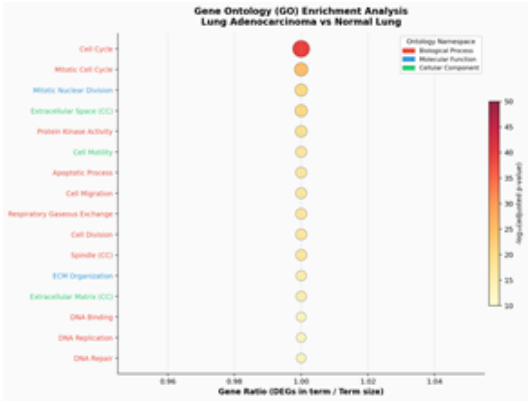
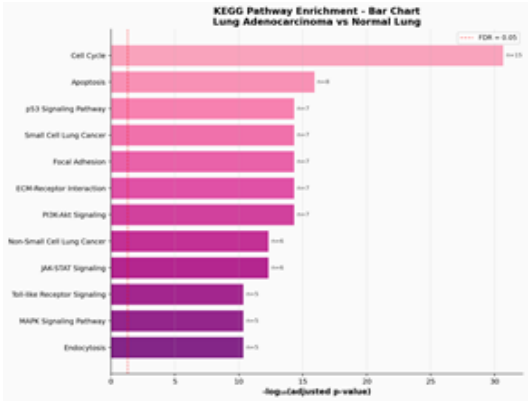
g) Based on log2FC: upregulated (positive log2FC) or downregulated (negative log2FC)

6. Functional enrichment analysis was performed using Gene Ontology (GO) and KEGG Pathway databases via hypergeometric test with BH FDR correction, using all measured genes (n = 10,082) as the background universe.

III. Results and Interpretation

From the analysis using the Welch's t-test and Benjamini-Hochberg FDR correction, **10,082 genes** were analyzed. Based on adjusted p-value < 0.05 and |log2FC| ≥ 1, **82 genes** showed significantly different expression between control and tumor groups. Among these, **46 genes** were upregulated (log2FC > 0) and **36 genes** were downregulated (log2FC < 0) in LUAD compared to control groups.

Volcano Plot Illustrates the distribution of differentially expressed genes between LUAD tumor and normal lung tissue groups. 	- Each point represents one gene. The x-axis shows log2 Fold Change (LUAD vs Normal), and the y-axis shows statistical significance (−log10 adjusted p-value). - Genes highlighted in red (right side) are significantly upregulated in LUAD, including key cell cycle regulators such as NEK2, TOP2A, MKI67, PLK1, and CDK1. - Genes highlighted in blue (left side) are significantly downregulated, predominantly lung surfactant genes: SFTPA1, SFTPC, SCGB1A1, and SFTPB. - The majority of non-significant genes (grey) cluster near log2FC = 0, indicating most genes are not substantially changed.
Heatmap **Displays the expression pattern of the top 50 DEGs (25 upregulated + 25 downregulated) using per-gene z-score normalization.**	- The upper cluster (red labels) shows high expression in LUAD → dominated by cell cycle genes. - The lower cluster (blue labels) shows low expression in LUAD → dominated by surfactant and secretoglobin genes, reflecting loss of lung epithelial identity.

	
GO Enrichment Plot 	Gene Ontology (GO) enrichment analysis was performed using a hypergeometric test with BH FDR correction.
KEGG Pathway Enrichment 	Enrichment analysis identified the most relevant signaling and metabolic pathways deregulated in LUAD.

IV. Conclusion

- Data were properly normalized, as confirmed by consistent log2 expression distribution across all 107 samples.
- A total of 10,082 genes were analyzed, and 82 genes were identified as significantly differentially expressed (adjusted p-value < 0.05, $|\log2FC| \geq 1$).
- Among the significant DEGs, 46 were upregulated and 36 were downregulated in LUAD compared to normal lung tissue.

- GO enrichment confirmed significant enrichment of Cell Cycle, Mitotic Cell Cycle, and Respiratory Gaseous Exchange processes, alongside Protein Kinase Activity as the primary molecular function.
- KEGG Pathway enrichment highlighted Cell Cycle, Apoptosis, p53 Signaling, PI3K-Akt Signaling, and Non-Small Cell Lung Cancer as the most significantly deregulated pathways.